

Machine Learning: The No-Nonsense (But Fun) Guide

Day 17: PCA – The "Find the Essence" Algorithm

Introduction

Imagine you're a photographer with 1,000 photos of a city, but you can only show 2 photos to someone.

Which ones would you choose?

- One photo showing the tall buildings (the skyline)
- One photo showing the busy streets (the city's energy)

Those two photos capture the essence of the city without showing every single detail.

Principal Component Analysis (PCA) works the same way with data. Instead of keeping every feature, it finds the most important ones that preserve the maximum information.

PCA: When you have 1,000 features but only need the few that matter most.

1. What is PCA?

PCA (Principal Component Analysis) is a dimensionality reduction technique that transforms high-dimensional data into a smaller set of new features called Principal Components, while preserving as much information (variance) as possible.

Before PCA	After PCA
1000 Features	2–10 Features
Difficult to visualize	Easy to visualize
Redundant information	Reduced redundancy
Noisy data	Cleaner data
Slower algorithms	Faster algorithms

2. Core Idea Behind PCA

Many features in a dataset are correlated.

Examples:

- Height and Weight
- Study Hours and Grades
- House Size and Price

Instead of using all correlated features, PCA creates new independent axes that capture the maximum variation in the data.

These new axes are called Principal Components.

Properties of Principal Components:

- They are uncorrelated with each other.
 - They are ordered by importance.
 - The first component captures the highest variance.
 - The second captures the next highest variance while remaining perpendicular to the first.
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3. How PCA Works

Step 1: Center the Data

The mean of every feature is subtracted so that the data is centered around zero.

Example:

Before:

10, 20, 30, 40

Mean = 25

After Centering:

-15, -5, 5, 15

Step 2: Find the Directions of Maximum Variance

PCA searches for directions where the data spreads the most.

- PC1 (First Principal Component): Captures maximum variance.
 - PC2 (Second Principal Component): Captures the next highest variance while remaining perpendicular to PC1.
 - Additional components continue similarly.
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Step 3: Project the Data

Instead of keeping every feature, PCA projects the data onto the top principal components.

Example:

Original Dataset

1000 Features

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PCA

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2 Principal Components

This makes visualization and analysis significantly easier.

4. Explained Variance

Explained variance tells us how much information is retained by each principal component.

Principal Component	Explained Variance	Cumulative Variance
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PC1	45%	45%
PC2	35%	80%
PC3	12%	92%
PC4	5%	97%

Rule of Thumb

Keep enough principal components to preserve 90–95% of the total variance.

5. Benefits of PCA

- Reduces dimensionality
 - Removes redundant information
 - Speeds up machine learning algorithms
 - Reduces noise
 - Helps visualize high-dimensional datasets
 - Reduces multicollinearity
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6. When to Use PCA

Use PCA When	Avoid PCA When
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Visualizing high-dimensional data	Features are already independent
Reducing noise	Feature interpretability is critical
Speeding up algorithms	Data contains highly non-linear relationships
Compressing data	Dataset already has very few features
Reducing multicollinearity	Individual feature importance must be preserved
Preprocessing before clustering	Linear assumptions do not hold

7. Common Problems and Solutions

Problem	Cause	Solution
First feature dominates	Data not scaled	Apply StandardScaler before PCA

Accuracy drops	Too few components retained	Increase the number of components
PCA doesn't separate data well	Non-linear relationships	Use t-SNE, UMAP, or Kernel PCA
Components are skewed	Presence of outliers	Remove or treat outliers before PCA

9. Real-World Applications

Domain	Application
Computer Vision	Image Compression
Face Recognition	Eigenfaces
Finance	Reducing correlated financial indicators
Bioinformatics	Gene expression analysis
Text Analytics	Topic modeling and document clustering

Anomaly Detection	Detecting unusual observations
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10. Interview Questions

- 1. What is PCA?**
 - 2. What problem does PCA solve?**
 - 3. What are Principal Components?**
 - 4. Why should data be standardized before PCA?**
 - 5. What is Explained Variance Ratio?**
 - 6. How do you choose the number of principal components?**
 - 7. What is the difference between PCA and t-SNE?**
 - 8. When does PCA perform poorly?**
 - 9. What is the Curse of Dimensionality?**
 - 10. What is the difference between Feature Selection and Feature Extraction?**
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Happy Learning !!!